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Data Science

AI-Assisted Document Intelligence Using Hybrid NLP and Large Language Models

BE_DS_A_Group No: 02

Name of The Guide:

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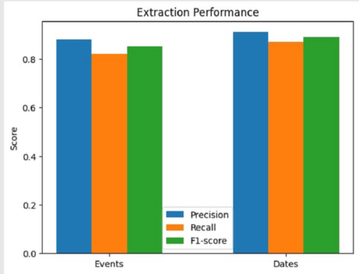
1. Problem Statement:

Legal processes are often complex and inefficient due to manual drafting, unclear user requirements, and heavy legal jargon. Users face difficulties in understanding responses, identifying risks, and managing multi-format documents. Additionally, poor organization and missed legal dates further reduce efficiency. These challenges highlight the need for an intelligent system to simplify contract generation, improve risk detection, enable conversational assistance, and organize legal timelines effectively.

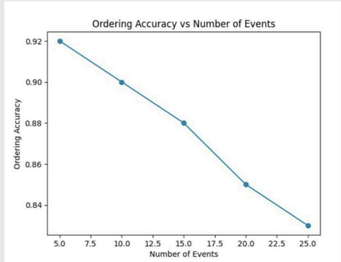
2. Objectives:

- Generate accurate contracts from user input
- Detect risks in uploaded documents
- Provide clear explanations and solutions
- Enable chatbot-based legal assistance
- Create automated legal timelines

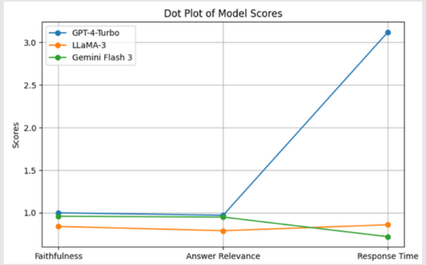
3. Results:



Performance Evaluation of Event and Date Extraction (Precision, Recall, F1-Score)

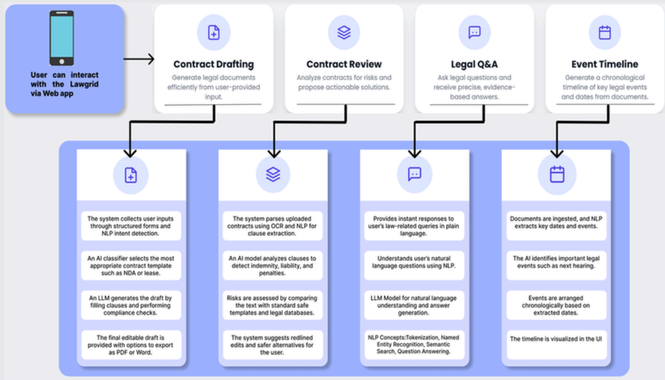


Impact of Event Count on Timeline Ordering Accuracy



Comparative Analysis of AI Models Based on Accuracy and Response Time

4. System Architecture:



5. References:

- P. Vimala Imogen, J. Sreenidhi, V. Nivedha, "AI-Powered Legal Docu-mentation Assistant", ISSN: 2582-2012, 2024, ResearchGate.
- Sweta Priyadarshni, "AI-Driven Document Automation and Compliance in Contract Life cycle Management", 2024, IEEE.

6. Conclusion:

LawGrid simplifies legal workflows using AI by automating contracts, detecting risks, and providing smart assistance. It improves efficiency, accuracy, and accessibility, making legal processes faster and more user-friendly.